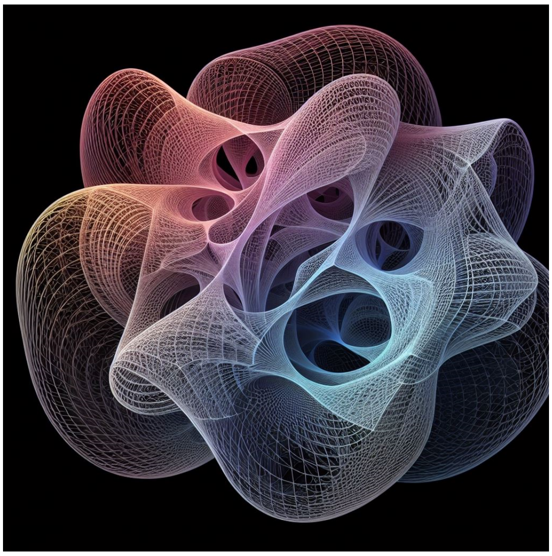

LINMA2474 - High-Dimensional Data Analysis and Optimization

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This summary may not be up-to-date, the newer version is available at this address: https://github.com/SimonDesmidt/Syntheses

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UCLouvain

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Optimization on manifolds

1.1 Introduction

Classical optimization methods like the gradient descent solve problems of the form

$$\min_{x \in \mathcal{M}} f(x) \tag{1.1}$$

for a set M. The methods rely on two key properties:

- Linearity: x_k and $\nabla f(x_k)$ belong to some vector space, in which they can be combined with linear operations;
- Inner product: $\nabla f(x_k)$ is the unique element of $\mathbb{R}^D$ such that

  $$\forall v \in \mathbb{R}^D,\ Df(x)[v] = \langle v, \nabla f(x) \rangle \tag{1.2}$$

  where $Df(x)[v] = \lim_{t \to 0} \frac{f(x+tv)-f(x)}{t}$ is the directional derivative of f at x in the direction v.

There are two ways to see the problem 1.1: as a constrained optimization problem, or as an unconstrained optimization problem assuming that nothing else exists outside the set $\mathcal{M}$. Optimization on manifolds extends the classical unconstrained optimization algorithms to problems whose search space is a manifold (will be defined later).

1.2 Definitions

1.2.1 Definition and properties of a manifold

Definition 1.1 (Optimisation on manifolds). To minimize a function f on a manifold $\mathcal{M}$, we need several objects:

- A local defining function $h : \mathbb{R}^d \to \mathbb{R}$ such that $h^{-1}(0) = \mathcal{M}$;
- The tangent space of $\mathcal{M}$ at some point x is the local linearization of $\mathcal{M}$ at $x \in \mathcal{M}$: u $\in$tangent space of $\mathcal{M}$ at x iff $u \in Ker(Dh(x))$;
- An inner product on the tangent spaces to define a new notion of gradient:

  $$Df(x)[v] = \langle \nabla f(x), v \rangle$$

- A retraction function, i.e. a tool that allows to make a step on a manifold in a given tangent direction.

Definition 1.2 (Smoothness). $F : \mathcal{U} \subseteq \mathcal{E} \to \mathcal{V} \subseteq \mathcal{E}'$, with $\mathcal{U}, \mathcal{V}$ open, is said to be smooth if it is $\mathcal{C}^\infty$ on its domain.

Let us explain the concepts needed for our optimization:

Definition 1.3 (Embedded submanifold and local defining function). Let $\mathcal{E}$ be a linear space of dimension d. A non-empty subset $\mathcal{M}$ of $\mathcal{E}$ is a smooth embedded submanifold of $\mathcal{E}$ of dimension n if either:

- $n = d$ and $\mathcal{M}$ is open in $\mathcal{E}$ (open submanifold);
- $n = d - k$ for some $k \geq 1$ and, for each $x \in \mathcal{M}$, there exists a neighbourhood $\mathcal{U}$ of x in $\mathcal{E}$ and a smooth function $h : \mathcal{U} \to \mathbb{R}^k$. In that case,
  - $\mathcal{M} \cap \mathcal{U} = h^{-1}(0) = \{y \in \mathcal{U} : h(y) = 0\}$ and
  - $rank(Dh(x)) = k$.

Such a function h is called a local defining function for $\mathcal{M}$ at x.

Definition 1.4 (Tangent space). Let $\mathcal{M}$ be a subset of $\mathcal{E}$. For all $x \in \mathcal{M}$, define

$$\mathcal{T}_x\mathcal{M} = \{c'(0) \mid c : \mathcal{I} \to \mathcal{M} \text{ is smooth and } c(0) = x\} \tag{1.3}$$

where $\mathcal{I}$ is any open interval containing $t = 0$. That means that v is in $\mathcal{T}_x\mathcal{M}$ iff there exists a smooth curve on $\mathcal{M}$ passing through x with velocity v.

Consider $\mathcal{M}$ an embedded submanifold of $\mathcal{E}$, $x \in \mathcal{M}$ and the set $\mathcal{T}_x\mathcal{M}$.

- If $\mathcal{M}$ is an open submanifold of $\mathcal{E}$, then $\mathcal{T}_x\mathcal{M} = \mathcal{E}$;
- Otherwise, $\mathcal{T}_x\mathcal{M} = Ker(Dh(x))$ with h any local defining function at x.

Definition 1.5 (Tangent bundle). The tangent bundle is the set of all tangent spaces: $\mathcal{T}\mathcal{M} = \{(x, v) : v \in \mathcal{T}_x\mathcal{M}\}$.

Definition 1.6 (Map between manifolds). Let $\mathcal{M}$ and $\mathcal{M}'$ be embedded submanifolds of $\mathcal{E}$ and $\mathcal{E}'$. A map $F : \mathcal{M} \to \mathcal{M}'$ is smooth iff $F = \bar{F}|_{\mathcal{M}}$ where $\bar{F}$ is some smooth map from a neighbourhood of $\mathcal{M}$ in $\mathcal{E}$ to $\mathcal{E}'$.

Definition 1.7 (Differential of a map between manifolds). The differential of $F : \mathcal{M} \to \mathcal{M}'$ at the point $x \in \mathcal{M}$ is the linear map $DF(x) : \mathcal{T}_x\mathcal{M} \to \mathcal{T}_{F(x)}\mathcal{M}'$ defined by

$$DF(x)[v] = \frac{d}{dt}F(c(t))|_{t=0} = (F \circ c)'(0) \tag{1.4}$$

where c is some smooth curve on $\mathcal{M}$ passing through x at $t = 0$ with velocity $v \in \mathcal{T}_x\mathcal{M}$.

$\rightarrow$ Note: the definition does not depend on the choice of the curve c: $DF(x) = D\bar{F}(x)|_{\mathcal{T}_x\mathcal{M}}$.

Definition 1.8 (**Retraction function**)**.** A retraction on a manifold $\mathcal{M}$ is a smooth map $R : \mathcal{TM} \to \mathcal{M} : (x, v) \to R_x(v)$ such that each curve $c(t) = R_x(tv)$ satisfies $c(0) = x$ and $c'(0) = v$.

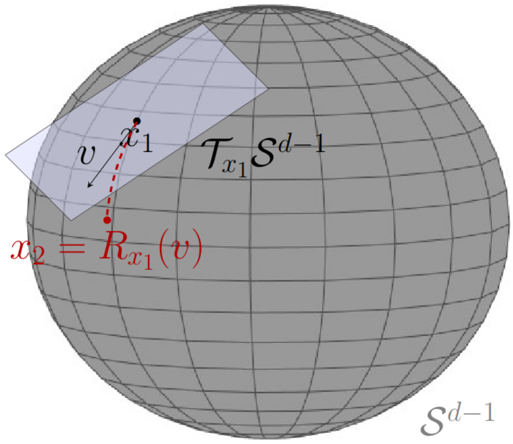

Figure 1.1: Retraction on a sphere.

1.2.2 Riemannian manifolds and metrics

Definition 1.9 (**Inner product**)**.** As seen in previous courses, an inner product on $\mathcal{T}_x\mathcal{M}$ is a bilinear, symmetric, positive definite function $\langle \cdot, \cdot \rangle_x : \mathcal{T}_x\mathcal{M} \times \mathcal{T}_x\mathcal{M} \to \mathbb{R}$. Note that the inner product depends on the point of linearization. It induces some norm for tangent vectors: $\|u\|_x = \sqrt{\langle u, u \rangle_x}$. A metric on $\mathcal{M}$ is a choice of inner product $\langle \cdot, \cdot \rangle_x$ for each $\mathcal{M}$.

Definition 1.10 (**Metric**)**.** A metric $x \to \langle \cdot, \cdot \rangle_x$ on $\mathcal{M}$ is a Riemannian metric if it varies smoothly with x, i.e. for all smooth vector fields V, W on $\mathcal{M}$, the function $x \to \langle V(x), W(x) \rangle_x$ is smooth from $\mathcal{M}$ to $\mathbb{R}$.

Definition 1.11 (**Riemannian manifold**)**.** A Riemannian manifold is a manifold with a Riemannian metric.

Definition 1.12 (**Riemannian distance**)**.** Let $\mathcal{M}$ be a Riemannian manifold. Given a smooth curve $c : [a, b] \to \mathcal{M}$, we define the length of c as

$$L(c) = \int_a^b \|c'(t)\|_{c(t)} dt \tag{1.5}$$

The Riemannian distance is then defined as $dist(x, y) = \inf_c L(c)$.

Definition 1.13 (**Riemannian submanifolds**)**.** Let $\mathcal{M}$ be an embedded submanifold of a Euclidean space $\mathcal{E}$. Equipped with the Riemannian metric obtained by restriction of the metric of $\mathcal{E}$, we call $\mathcal{M}$ a Riemannian submanifold of $\mathcal{E}$.

1.2.3 Gradient on manifolds

Definition 1.14 (Riemannian gradient). Let $f : \mathcal{M} \to \mathbb{R}$ be smooth on a Riemannian manifold $\mathcal{M}$. The Riemannian gradient of f is the vector field $\mathrm{grad} f$ on $\mathcal{M}$ uniquely defined by the following identities:

$$\forall (x, v) \in \mathcal{TM}, \qquad Df(x)[v] = \langle v, \mathrm{grad} f(x) \rangle_x \tag{1.6}$$

where $Df(x)$ is the differential of f and $\langle \cdot, \cdot \rangle_x$ is the Riemannian metric.

Theorem 1.15. Let $\mathcal{M}$ be a Riemannian submanifold of $\mathcal{E}$ endowed with the metric $\langle \cdot, \cdot \rangle$ and let $f : \mathcal{M} \to \mathbb{R}$ be a smooth function. The Riemannian gradient of f is given by

$$\mathrm{grad} f(x) = \mathrm{Proj}_x(\nabla \bar{f}(x)) \tag{1.7}$$

where $\bar{f}$ is any smooth extension of f to a neighborhood of $\mathcal{M}$ in $\mathcal{E}$, and $\nabla \bar{f}(x)$ is the Euclidean gradient of $\bar{f}$ at x.

→ Note: for $\mathcal{E} = \mathbb{R}^d$ and using the usual metric $\langle u, v \rangle = u^T v$, the projection operator is

$$\mathrm{Proj}_x(v) = v - (x^T v)x \tag{1.8}$$

Proposition 1.16. Let $f : \mathcal{M}_1 \to \mathcal{M}_2$ and $G : \mathcal{M}_2 \to \mathcal{M}_3$ be smooth, where $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$ are embedded submanifolds of $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3$ respectively. Then

$$G \circ F : \mathcal{M}_1 \to \mathcal{M}_3 : x \to G(F(x)) \tag{1.9}$$

is smooth and the chain rule applies:

$$D(G \circ F)(x)[v] = DG(F(x))[DF(x)[v]] \tag{1.10}$$

1.2.4 Taylor development of functions defined on manifolds

Let $f : \mathcal{M} \to \mathbb{R}$ be smooth and $c : \mathcal{I} \to \mathcal{M}$ be a smooth curve with $c(0) = x$ and $c'(0) = v$, with $\mathcal{I} \subseteq \mathbb{R}$ an open interval around $t = 0$ and $\|v\|_x = 1$. Let us define $g : \mathcal{I} \to \mathbb{R} : t \to g(t) = f(c(t))$. Since $g = f \circ c$ is smooth and maps real numbers to real numbers, it admits a Taylor expansion:

$$g(t) = g(0) + tg'(0) + \mathcal{O}(t^2) \tag{1.11}$$

By the chain rule,

$$g'(t) = Df(c(t))[c'(t)] = \langle \mathrm{grad} f(c(t)), c'(t) \rangle_{c(t)} \tag{1.12}$$

and for $t = 0$,

$$g(0) = f(x) \qquad g'(0) = \langle \mathrm{grad} f(x), v \rangle_x \tag{1.13}$$

Therefore,

$$\begin{aligned} f(c(t)) &= f(x) + t\langle \mathrm{grad} f(x), v \rangle_x + \mathcal{O}(t^2) \\ f(R_x(tv)) &= f(x) + \langle \mathrm{grad} f(x), tv \rangle_x + \mathcal{O}(t^2) \end{aligned} \tag{1.14}$$

And defining $s := tv \in \mathcal{T}_x\mathcal{M}$,

$$f(R_x(s)) = f(x) + \langle \mathrm{grad} f(x), s \rangle_x + \mathcal{O}(\|s\|_x^2) \tag{1.15}$$

This allows to define the Riemannian gradient descent in the next chapter (see section 2.2).

1.3 Geometric considerations

1.3.1 Geodesics

Definition 1.17 (Geodesics). On a Riemannian manifold $\mathcal{M}$, a geodesic is a smooth curve $c : \mathcal{I} \to \mathcal{M}$ such that $c''(t) = 0$ for all $t \in \mathcal{I}$, where $\mathcal{I}$ is an open interval of $\mathbb{R}$. The acceleration $c''(t)$ on a Riemannian submanifold is defined as follows:

$$c''(t) = \mathrm{Proj}_{c(t)}(\ddot{c}(t)) \qquad \text{where} \qquad \ddot{c}(t) := \frac{d^2}{dt^2}c(t) \tag{1.16}$$

For example, the curve $c(t) = \cos(t\|v\|)x + \frac{\sin(t\|v\|)}{\|v\|}v$, where x is the starting point and v the initial velocity ($v \in \mathcal{T}_x\mathcal{S}^{d-1}$), i.e. the curve that traces a great circle on the sphere.

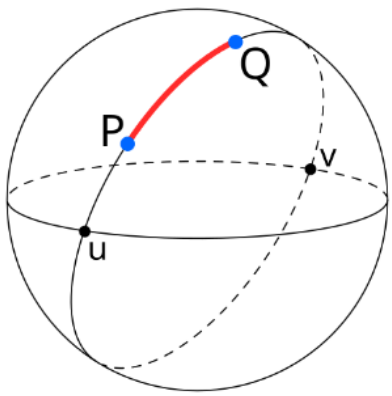

Figure 1.2: Example of geodesic on the sphere.

Theorem 1.18. Every geodesic γ on $\mathcal{M}$ is locally minimizing, i.e. every t in the domain of γ has a neighborhood $\mathcal{I}$ in the domain of γ such that, if $a, b \in \mathcal{I}$ with $a < b$, then the restriction $\gamma|_{[a,b]}$ is a minimizing curve.

→ Note: not all geodesics are minimizing, e.g. on the sphere, two points are always linked by two geodesics, a long one and a short one (red and black curve on figure 1.2).

Theorem 1.19. On a Riemannian manifold, for every $(x, v) \in \mathcal{TM}$, there exists a unique maximal geodesic $\gamma : \mathcal{I} \to \mathcal{M}$ with $\gamma(0) = x$ and $\gamma'(0) = v$, where maximal means that the interval $\mathcal{I}$ is as large as possible. This is equivalent to a Cauchy problem:

$$\gamma''(t) = 0 \qquad \text{such that} \begin{cases} \gamma(0) = x \\ \gamma'(0) = v \end{cases} \tag{1.17}$$

Definition 1.20 (Exponential map). Consider the set

$$\mathcal{D} = \{(x, v) \in \mathcal{TM} \ : \gamma \text{ is defined on an interval containing } [0, 1]\}$$

which is a subset of the tangent bundle. The exponential map $Exp : \mathcal{D} \to \mathcal{M}$ is defined by the relation $Exp(x, v) = \gamma(1)$.

→ Note: the exponential map is a retraction.

1.3.2 Geodesic convexity

As a reminder, in linear space, a set $\mathcal{S}$ is convex if, for all $x, y \in \mathcal{S}$, the line segment $t \to (1-t)x + ty$ for $t \in [0,1]$ is also in $\mathcal{S}$. A function $f : \mathcal{S} \to \mathbb{R}$ is a convex function if $\mathcal{S}$ is convex and for all $x, y \in \mathcal{S}$ we have

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y) \qquad \forall t \in [0,1] \tag{1.18}$$

Definition 1.21 (Geodesic convexity). A subset $\mathcal{S}$ of a Riemannian manifold $\mathcal{M}$ is geodesically convex if, for every $x, y \in \mathcal{S}$, there exists a geodesic segment $c : [0,1] \to \mathcal{M}$ such that $c(0) = x$, $c(1) = y$ and $c(t)$ is in $\mathcal{S}$ for all $t \in [0,1]$.

Definition 1.22 (Geodesically convex function). A function $f : \mathcal{S} \to \mathbb{R}$ is geodesically convex if $\mathcal{S}$ is geodesically convex and $f \circ c : [0,1] \to \mathbb{R}$ is convex for each geodesic segment $c : [0,1] \to \mathcal{M}$ whose image is in $\mathcal{S}$.

Theorem 1.23. Let $\mathcal{S}$ be a geodesically convex set on a Riemannian manifold $\mathcal{M}$ and let $f : \mathcal{M} \to \mathbb{R}$ be differentiable in a neighborhood of $\mathcal{S}$. Then, the restriction $f|_{\mathcal{S}} : \mathcal{S} \to \mathbb{R}$ is geodesically convex iff for all geodesic segments $c : [0,1] \to \mathcal{M}$ contained in $\mathcal{S}$, we have (for $x = c(0)$)

$$f(c(t)) \geq f(x) + t\langle \operatorname{grad} f(x), c'(0) \rangle_x \qquad \forall t \in [0,1] \tag{1.19}$$

Theorem 1.24. If f is differentiable and geodesically convex on an open geodesically convex set, then x is a global minimizer of f iff $\operatorname{grad} f(x) = 0$.

1.4 Euclidean and Non-Euclidean geometries

In general, in geometry, we say that two lines are parallel iff they are disjoint. In Euclidean geometry, if L is a line and if P is a point not in L, then there exists one and only one line through P that is parallel to L. However, in a hyperbolic geometry, there exist infinitely many distinct hyperbolic lines through P that are parallel to L. Conversely, in elliptic geometries, there exist no line through P that is parallel to L.

The curvature of a surface is defined by the biggest circle that stays on one side of the curve.

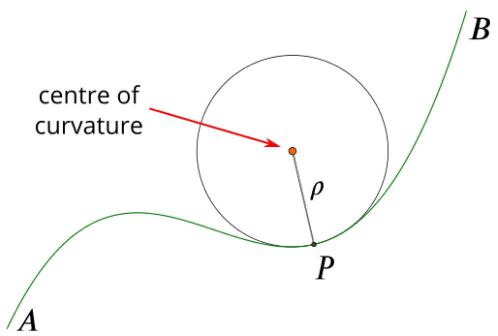

Figure 1.3: Illustration of the curvature and centre of curvature for a function.

The directions in the normal plane where the curvature takes its maximum and minimum values (k_1 and k_2) are always perpendicular and are called principal directions. The product $k_1 k_2$ of the two principal curvatures is called the Gaussian curvature.

Riemannian gradient descent

2.1 Topology tools

2.1.1 General topology

Definition 2.1 (Topology). A topology on a set X is a collection T of subsets of X, called open sets, such that

- X and $\emptyset$ belong to X;
- the union of the elements of any subcollection of T is in T;
- the intersection of the elements of any finite subcollection of T is in T.

Definition 2.2 (Hausdorff topological space.). A topological space is a couple (X, T) where X is a set and T is a topology on X. The topological space (X, T) is Hausdorff if any two distinct points of X have disjoint neighborhoods. If X is Hausdorff, then every sequence of points of X converges to at most one point in X.

2.1.2 Topology for embedded submanifolds

Definition 2.3 (Open set.). A subset $\mathcal{U}$ of $\mathcal{M}$ is open (respectively closed) if it is the intersection of $\mathcal{M}$ with an open (respectively closed) set of $\mathcal{E}$.

Definition 2.4 (Neighborhood.). A neighborhood of x in $\mathcal{M}$ is an open subset of $\mathcal{M}$ that contains x.
A neighborhood of a subset of $\mathcal{M}$ is an open subset of $\mathcal{M}$ that contains that subset.

Definition 2.5 (Limit.). Consider a sequence s of points $x_0, x_1, x_2, \ldots$ on a manifold $\mathcal{M}$.

- We say that a point x in $\mathcal{M}$ is a limit of s if, for all neighborhood $\mathcal{U}$ of x, there exists some $K \in \mathbb{Z}$ such that x_k is in $\mathcal{U}$ for all $k \geq K$.
  The topology of a manifold is Hausdorff, hence a sequence has at most one limit. If x is the limit, we write
  $$\lim_{k\to\infty} x_k = x \qquad \text{or} \qquad x_k \to x \tag{2.1}$$
  and we say that the sequence converges to x.
- A point $x \in \mathcal{M}$ is an accumulation point of s if it is the limit of a subsequence of s.

2.2 Riemannian gradient descent algorithm

The basic algorithm is the following:

Algorithm 1 Riemannian gradient descent

1: **Input:** $x_0 \in \mathcal{M}$;
2: **for** $k = 0, 1, 2, \dots$ **do**
3: Pick a step size $\alpha_k > 0$
4: $x_{k+1} = R_{x_k}(-\alpha_k \text{grad} f(x_k))$
5: **end for**

There are several ways to choose the step size α_k:

- a fixed step size: $\alpha_k = \alpha$ for all k;
- optimal step size: compute α_k that minimizes exactly the function

$$g(\alpha) = f\left(R_{x_k}(-\alpha \text{grad} f(x_k))\right) \tag{2.2}$$

- Backtracking: starting with a guess $\alpha_0 > 0$, iteratively reduce it by a factor $\tau \in (0, 1)$ until it is deemed acceptable.

2.2.1 Global convergence

Let $f : \mathcal{M} \to \mathbb{R}$ be a smooth function on a Riemannian manifold $\mathcal{M}$, and assume that

- there exists a lower bound $f_{low} \in \mathbb{R}$ such that $f(x) \geq f_{low}$ for all $x \in \mathcal{M}$;
- there exists a constant $c > 0$ such that, for all $k \in \mathbb{Z}$,

$$f(x_k) - f(x_{k+1}) \geq c \|\text{grad} f(x_k)\|_{x_k}^2 \tag{2.3}$$

Then,

$$\lim_{k \to \infty} \|\text{grad} f(x_k)\|_{x_k}^2 = 0 \tag{2.4}$$

Furthermore, for all $K \geq 1$, there exists $k \in \{0, \dots, K-1\}$ such that

$$\|\text{grad} f(x_k)\|_{x_k} \leq \sqrt{\frac{f(x_0) - f_{low}}{cK}} \tag{2.5}$$

This means that if a precision ϵ is required on the solution, the number of iterations that we will need has a complexity $\mathcal{O}(\epsilon^{-2})$.

The proof of these results is the same as done in the courses LINMA2471 or LINMA2460.

2.2.2 Ensuring the assumptions

In classical optimisation, the Lipschitz-smoothness assumption is that there exists a constant $L > 0$ such that

$$\|\nabla f(x+s) - \nabla f(x)\| \leq L\|s\| \qquad \forall x, s \in \mathbb{R}^d \tag{2.6}$$

implied by the expression

$$f(x+s) \leq f(x) + \langle \nabla f(x), s \rangle + \frac{L}{2}\|s\|^2 \qquad \forall x, s \in \mathbb{R}^d \tag{2.7}$$

However, on manifolds, the gradient is defined differently and with a different support. We need to modify this condition.
On a manifold $\mathcal{M}$, for a given subset S of the tangent bundle $\mathcal{TM}$, there exists a constant $L > 0$ such that, forall $(x, s) \in S$,

$$f(R_x(s)) \leq f(x) + \langle \text{grad} f(x), s \rangle + \frac{L}{2}\|s\|_x^2 \tag{2.8}$$

From this, we can ensure the assumption of equation (2.3): define $x := x_k$, $s := -\alpha_k \text{grad} f(x_k)$ and $x_{k+1} := R_x(s)$. Then,

$$f(x_{k+1}) \leq f(x_k) - \alpha_k \left(1 - \frac{L\alpha_k}{2}\right) \|\text{grad} f(x_k)\|_{x_k}^2 \tag{2.9}$$

and this gives

$$f(x_k) - f(x_{k+1}) \geq \alpha_k \left(1 - \frac{L\alpha_k}{2}\right) \|\text{grad} f(x_k)\|_{x_k}^2 \tag{2.10}$$

Therefore, the assumption is true taking $c = \alpha_k \left(1 - \frac{L\alpha_k}{2}\right)$.

2.2.3 Optimal step size

From the previous proof, we know that the decrease of the iterate is best when the constant c is maximized. We can thus define the function

$$g : \mathbb{R} \to \mathbb{R} : \alpha \to \alpha \left(1 - \frac{L\alpha}{2}\right) \tag{2.11}$$

which is maximized for $\alpha^* = 1/L$, the same value seen for optimization in $\mathbb{R}^d$. For that value, the constant is $c = 1/2L$.

2.2.4 Backtracking

If the Lipschitz constant L is unknown, a possible algorithm for the optimal step size if backtracking line-search using the Armijo-Goldstein condition. This condition is

$$f(x) - f(R_x(-\alpha \text{grad} f(x))) \geq r\alpha \|\text{grad} f(x)\|_x^2 \tag{2.12}$$

and the algorithm is

Algorithm 2 Backtracking line-search

Parameters: $\tau, r \in (0,1)$, e.g. $\tau = 1/2$ and $r = 10^{-4}$;
Input: $x \in \mathcal{M}$ and $\bar{\alpha} > 0$;
Set $\alpha \leftarrow \bar{\alpha}$;
while $f(x) - f(R_x(-\alpha \mathrm{grad} f(x))) < r\alpha \|\mathrm{grad} f(x)\|_x^2$ **do**
    $\alpha \leftarrow \tau\alpha$
end while
return α

Under an assumption similar to the Lipschitz condition on the gradient, a convergence rate of $\mathcal{O}(1/\epsilon^2)$ can still be guaranteed.

PAA

3.1 Dynamical systems

Dynamical systems in physics can generally be modelled by a set of differential equations, giving a state-space model:

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= g(x(t), u(t)) \end{aligned} \tag{3.1}$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^{n_i}$ is the input and $y \in \mathbb{R}^{n_o}$ is the observed output. As their expressions are governed by the system, x and y cannot be free and are restricted to a low-dimensional submanifold in their respective functional space.

3.1.1 The discrete linear time invariant model

This model is

$$\begin{aligned} x(t+1) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{aligned} \tag{3.2}$$

Our goal is to determine the dimension n of the state-space model, and the matrix parameters A, B, C, D.
For that, let us define two Hankel matrices:

$$\begin{aligned} Y &= \begin{bmatrix} y(1) & y(2) & \dots & y(N) \\ y(2) & y(3) & \dots & y(N+1) \\ \vdots & \vdots & \ddots & \vdots \\ y(N) & y(N+1) & \dots & y(2N-1) \end{bmatrix} \in \mathbb{R}^{n_o N \times N} \\ U &= \begin{bmatrix} u(1) & u(2) & \dots & u(N) \\ u(2) & u(3) & \dots & u(N+1) \\ \vdots & \vdots & \ddots & \vdots \\ u(N) & u(N+1) & \dots & u(2N-1) \end{bmatrix} \in \mathbb{R}^{n_i N \times N} \end{aligned} \tag{3.3}$$

From the system, we know that

$$\begin{aligned} y(1) &= Cx(1) + Du(1) \\ y(2) &= Cx(2) + Du(2) = C(Ax(1) + Bu(1)) + Du(2) \\ &\vdots \end{aligned} \tag{3.4}$$

By this recurrence, we can see that

$$\begin{bmatrix} y(1) \\ \vdots \\ y(r) \end{bmatrix} = \underbrace{\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{r-1} \end{bmatrix}}_{=:G} x(1) + \underbrace{\begin{bmatrix} D & & & & \\ CB & D & & & \\ CAB & CB & D & & \\ \vdots & \ddots & \ddots & \ddots & \\ CA^{r-2}B & CA^{r-3}B & \dots & CB & D \end{bmatrix}}_{=:H} \begin{bmatrix} u(1) \\ \vdots \\ u(r) \end{bmatrix} \tag{3.5}$$

And thus

$$Y = GX + HU \tag{3.6}$$

where $X = [x(1), x(2), \dots, x(N)] \in \mathbb{R}^{n \times N}$. Let $U^{\perp}$ span the columns of the nullspace of U. Then, $UU^{\perp} = 0$, and

$$YU^{\perp} = GXU^{\perp} + HUU^{\perp} = GXU^{\perp} \tag{3.7}$$

We call $YU^{\perp}$ the input-output matrix, and this is always of rank lower than the dimension of the state space: $rank(YU^{\perp}) \leq n$.
Man-made structures and object often display patterns and/or symmetries in their making, meaning that their actual information space has a much lower dimensionality. Therefore, using the matrix $YU^{\perp}$ is much more efficient.

3.1.2 Discretize a continuous system

To discretize a function $x(t) : \mathbb{R} \to \mathbb{R}$ to a function $x_T(t) : \mathbb{N} \to \mathbb{R}$, we use the Sha function:

$$Ш(t) := \sum_{k=-\infty}^{\infty} \delta(t-k) \tag{3.8}$$

With that function, the discretization is straightforward:

$$x_T(t) = x(t) \cdot Ш\left(\frac{t}{T}\right) \tag{3.9}$$

And in the frequency domain:

$$\mathcal{F}(x_T(t)) = \hat{x}(\nu) \star TШ(t\nu) \tag{3.10}$$

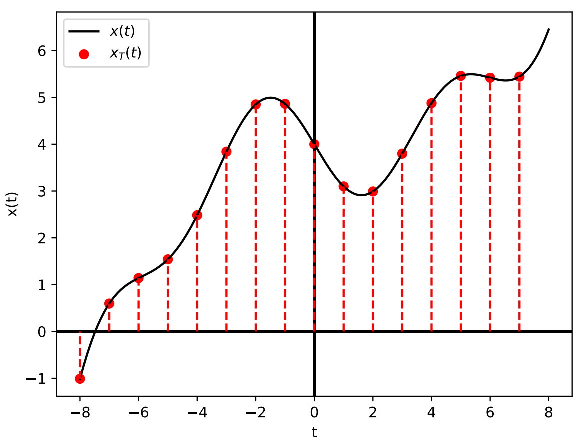

Figure 3.1: Discretization using the Sha function.

3.1.3 Latent variables

A set of random variables $x_o \in \mathbb{R}^{n_o}$ can be given by a probability distribution that depends on other hidden (or latent) variables $x_h \in \mathbb{R}^{n_h}$. In that case, the structure of the data is fully described by the joint distribution of the vector $x = (x_o, x_h) \in \mathbb{R}^n$, for $n = n_o + n_h$. Assuming that $x \sim \mathcal{N}(0, \Sigma)$, where $\Sigma \in \mathbb{R}^{n \times n}$ is the covariance matrix, we define $\Theta = \Sigma^{-1}$ the precision matrix, or concentration matrix. Let us divide the matrices in blocks:

$$\Sigma = \begin{bmatrix} \Sigma_o & \Sigma_{o,h} \\ \Sigma_{o,h}^T & \Sigma_h \end{bmatrix} = \begin{bmatrix} \Theta_o & \Theta_{o,h} \\ \Theta_{o,h}^T & \Theta_h \end{bmatrix}^{-1} \in \mathbb{R}^{n \times n} \tag{3.11}$$

As the matrices of hidden variables are by definition not observed, we need a tool to connect Σ_o to Θ_o. This tool is the Schur complement:

$$\Sigma_o^{-1} = \Theta_o - \Theta_{o,h} \Theta_h^{-1} \Theta_{o,h}^T \in \mathbb{R}^{n_o \times n_o} \tag{3.12}$$

If the observed random variables are mainly independent, then Θ_o is sparse. Moreover, the second term has a rank less than the number of hidden variables n_h, which itself is often relatively small compared to n_o. Therefore, the covariance matrix Σ_o of the observed variables is always a decomposable structure $\Sigma_o^{-1} = S + L$, where S is sparse and L is low-rank, two forms that are very useful in high-dimensional analysis. Earlier results allow us to write the conditional probabilities:

If $\begin{pmatrix} x_a \\ x_b \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_a \\ \mu_b \end{pmatrix}, \begin{pmatrix} \Sigma_a & \Sigma_{a,b} \\ \Sigma_{a,b} & \Sigma_b \end{pmatrix} \right)$, then

$$x_a | x_b \sim \mathcal{N}(\bar{\mu}, \bar{\Sigma}) \qquad \begin{cases} \bar{\mu} = \mu_a + \Sigma_{a,b} \Sigma_b^{-1} (x_b - \mu_b) \\ \bar{\Sigma} = \Sigma_a - \Sigma_{a,b} \Sigma_b^{-1} \Sigma_{a,b}^T \end{cases} \tag{3.13}$$

Je vois pas quoi écrire sur les sections 1.2 et 1.3. Demander à PAA.

3.2 Sparse signal modeling

Often, the number of unknowns is much larger than the number of observations: we have $y = Ax$, for $y \in \mathbb{R}^n$ the observations, $A \in \mathbb{R}^{m \times n}$ the data generation process and $x \in \mathbb{R}^n$ the unkowns. We consider $m \ll n$. The system thus have infinitely many solutions or none. To find the one solution that we want, we need to leverage some additional properties such as sparsity.

3.2.1 Example in medical imaging

Consider a digital $N \times N$ image $I \in \mathbb{R}^{N^2}$ (not the identity matrix!). This image, in an MRI, corresponds to the density of protons at a given spatial location in the brain. The input u denotes how the magnetic field we applied varies over space. Denote $\mathcal{F}$ the 2D Fourier transform, we have

$$y = \mathcal{F}[I](u) \tag{3.14}$$

By changing the applied magnetic field, u varies and we can collect m samples of the Fourier transform: $\{u_1, \dots, u_m\}$. This gives (3.14) in m dimensions. The system seems impossible to solve, but we can use a known structure about it: the wavelet transform.

$$I = \Psi x \tag{3.15}$$

where Ψ is a matrix defined from $\psi \in L^2(\mathbb{R})$, with

$$\Psi_{j,k}(x) := 2^{j/2}\psi(2^j x - k) \qquad x \in \mathbb{R} \tag{3.16}$$

The function ψ is said to be a wavelet if the system $\{\Psi_{j,k}\}_{j,k\in\mathbb{Z}}$ is an orthonormal basis of $L^2(\mathbb{R})$[1]. Moreover, $x \in \mathbb{R}^{N^2}$ containts the coefficients of the image I with respect to the basis Ψ. Many of these coefficients are extremely small, meaning that x is sparse. We can try to reconstruct this vector x by using the observation equation:

$$\begin{aligned} y = \mathcal{F}_U[I] &= \mathcal{F}_U\left[\psi_1 x_1 + \cdots + \psi_{N^2} x_{N^2}\right] \\ &= \mathcal{F}_U[\psi_1]x_1 + \cdots + \mathcal{F}_U(\psi_{N^2})x_{N^2} \\ &= \left[\mathcal{F}_U[\psi_1] \dots \mathcal{F}_U[\psi_{N^2}]x = Ax\right] \end{aligned} \tag{3.17}$$

for the matrix $A \in \mathbb{R}^{m\times N^2}$.
This system is much underdetermined, but we can use the sparsity of x to solve it: we consider that only its k largest coefficients are significant and the others are negligible. The solution $\hat{x}$ will be used to define an approximation of I:

$$\hat{I} = \sum_{i=1}^{N^2} \psi_i \hat{x}_i \tag{3.18}$$

3.2.2 Recovering a sparse solution

Recall the three properties of a norm in a space V:

- Nonnegative homogeneity: $\|\alpha x\| = |\alpha|\|x\|$ for all vector $x \in V$ and scalars $\alpha \in \mathbb{R}$;
- Positive definiteness: $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$;
- Subadditivity: $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in V$.

The ℓ_0 norm is the number of nonzero entries in a vector x:

$$\|x\|_0 = \#\{i | x(i) \neq 0\} \tag{3.19}$$

It is thus a measure of the sparsity of a vector. It is not a formal norm, as it does not verify the nonnegative homogeneity property, but it can be seen as a limit of the ℓ^p norm:

$$\|x\|_0 = \lim_{p\to 0^+} \|x\|_p^p \tag{3.20}$$

[1]The scalar product used is $\langle f, g\rangle =: \int_{-\infty}^{\infty} f(x)g(x)dx$

Minimizing the ℓ^0 norm

Knowing y and A, we want to find x_o by recovering the sparsest x such that $Ax = y$. Namely, we want to find

$$\hat{x} \in \arg\min \|x\|_0 \qquad \text{s.t. } Ax = y \tag{3.21}$$

Theorem 3.1. Assuming that $y = Ax_o$, $\|x_o\|_0 \leq k$, $\hat{x}$ exists and $\{\delta \in \mathbb{R}^n | A\delta = 0, \|\delta\|_0 \leq 2k\} = \{0\}$, i.e. the nullspace of A has no sparse vector, then $\hat{x} = x_o$ and is the unique solution of the ℓ^0 minimization.

$\rightarrow$ Note: the bound on the sparsity of δ is $2k$ because the norm of $\hat{x} - x_o$ is bounded by $2k$: $\|\hat{x} - x_o\|_0 \leq \|\hat{x}\|_0 + \|x_o\|_0 \leq 2k$ and $A\delta = A(\hat{x} - x_o) = A\hat{x} - Ax_o = y - y = 0$.

Definition 3.2. The Kruskal rank is defined as

$$krank(A) := \max_r \{r \in \mathbb{N} | \text{ the columns of } A_I \text{ are linearly independent } {\color{red}\forall} I \subset [n] \text{ with } |I| = r\} \tag{3.22}$$

The difference with the usual rank is the red $\forall$, that is a $\exists$ for the usual rank.

$\rightarrow$ Note: $0 \leq krank(A) \leq rank(A) \qquad \forall A$.

Theorem 3.3 (**ℓ^0 recovery.**)**.** Suppose that $y = Ax_o$, with $\|x_o\|_0 \leq \frac{1}{2} krank(A)$. Then, x_o is the unique optimal solution to the ℓ^0 minimization problem.

All these results make the ℓ^0 minimization problem well-posed from an initial ill-posed problem, but the worst-case complexity of an algorithm solving this problem is n^k, for k the number of nonzero entries that we want to recover: the problem is NP-hard.

3.2.3 Relaxing the sparse recovery problem

As the problem is NP-hard, we need a remplacement to make it easier to solve. We can see that in 1D, the absolute value function is the convex hull of the ℓ^0 norm on the interval $[-1, 1]$ (see figure **??**).

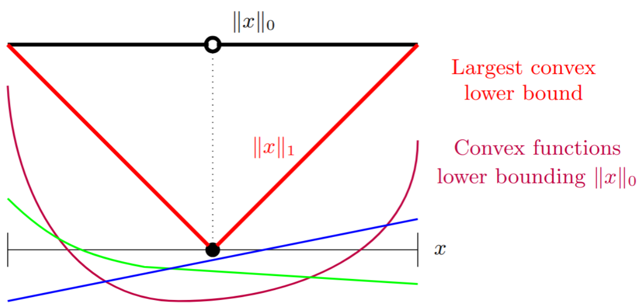

Figure 3.2: Replacing the ℓ^0 norm by its convex envelope.

Then, the ℓ^1 norm provides a good replacement for the ℓ^0 norm and simplifies the problem to a convex optimisation problem.

$$\min_x \|x\|_1 \qquad \text{such that } y = Ax \tag{3.23}$$

To solve this new problem, we can use a projected subgradient method:

$$x_{t+1} := P_{\mathcal{C}}\left(x_t - \frac{1}{t} sign(x_t)\right) \tag{3.24}$$

for $P_{\mathcal{C}}(z) = \arg\min_{x \in \mathcal{C}} \frac{1}{2}\|z - x\|_2^2$. Here, we use $\mathcal{C} = \{x : Ax = y\}$ and the sign function is the subgradient of the ℓ^1 norm.
If A has full row rank (plausible in the general case), then the projection is easy:

$$\mathcal{P}_{\{x|Ax=y\}}(z) = z - A^T(AA^T)^{-1}(Az - y) \tag{3.25}$$

3.3 Incoherent matrices

Definition 3.4. The **mutual coherence** of a matrix $A = [a_1|\cdots|a_n] \in \mathbb{R}^{m\times n}$ with all nonzero columns is defined as:

$$\mu(A) = \max_{i\neq j} \left| \langle \frac{a_i}{\|a_i\|_2}, \frac{a_j}{\|a_j\|_2} \rangle \right| \tag{3.26}$$

The mutual coherence is useful for sparse recovery because we will show that when $\mu(A)$ is small, the $krank(A)$ will be large. This is because if the columns of A are close to orthogonal, then any subset of them will be linearly independent and it will be well-conditioned as their smallest singular value $\sigma_{\min}$ will be close to their largest singular value $\sigma_{\max}$. Consider that A_I is the full-rank submatrix of A formed by the columns indexed by $I \subset [n]$ with $|I| = k$. To prove that we want to write the matrix under a form that is the sum of the diagonal and off-diagonal (Δ) entries:

$$A_I^* A_I = I + \Delta \tag{3.27}$$

We can bound the spectral norm of Δ by its Frobenius norm, then by the max norm, and finally by the mutual coherence:

$$\|\Delta\|_{SP} \leq \|\Delta\|_F < k\|\Delta\|_{l_\infty} \leq k\mu(A) \tag{3.28}$$

Then using this result, we can bound the smallest singular value of $A_I^* A_I$. Let $\Delta = UDU^T$ be an EVD then $\|\Delta\|_{SP} = \max_i |d_i|$:

$$\begin{aligned}
\sigma_{\min}(A_I^* A_I) &= \sigma_{\min}(I + \Delta) = \sigma_{\min}(U(I + D)U^T) \\
&= \min_i |1 + d_i| \geq \min_i 1 - |d_i| \\
&= 1 - \max_i |d_i| = 1 - \|\Delta\|_{SP} \\
&> 1 - k\mu(A)
\end{aligned} \tag{3.29}$$

Then we have:

$$1 - k\mu(A) < \sigma_{\min}(A_I^* A_I) \leq \sigma_{\max}(A_I^* A_I) < 1 + k\mu(A) \tag{3.30}$$

With the previous result, we can see that if $1 - k\mu(A) \geq 0$ or equivalently $k \leq 1/\mu(A)$, then $\sigma_{\min}(A_I^* A_I) > 0$ and thus A_I is full rank.
Combining these results with the definition of krank, we have the following result:

$$krank(A) \geq \frac{1}{\mu(A)} \tag{3.31}$$

This can be proven like this:
Let $s := krank(A) + 1$. We normalize the columns of A, this does not change the *krank* nor the mutual coherence. Then $\exists a_{j_1}, \ldots, a_{j_s}$ linearly dependent, and thus $\alpha_1 a_{j_1} + \cdots + \alpha_s a_{j_s} = 0$ for some $\alpha_i \neq 0$ and $\alpha_1 = \max_i |\alpha_i|$. Then, we have

$$\begin{aligned}
0 &= \langle a_{j_1}, \alpha_1 a_{j_1} + \cdots + \alpha_s a_{j_s} \rangle \\
\alpha_1 &= -\alpha_2 \langle a_{j_1}, a_{j_2} \rangle - \cdots - \alpha_s \langle a_{j_1}, a_{j_s} \rangle \\
\frac{\alpha_1}{|\alpha_1|} &= \left(\frac{|\alpha_2|}{|\alpha_1|} + \cdots + \frac{|\alpha_s|}{|\alpha_1|} \right) \mu(A) \\
1 &\leq (s-1)\mu(A) \\
\frac{1}{\mu(A)} &\leq krank(A)
\end{aligned} \tag{3.32}$$

With this result, we can see that if $y = Ax_0$ and $\|x_o\|_0 \leq \frac{1}{2\mu(A)}$ then x_o is the unique optimal solution to the ℓ^0 minimization problem.

$$\min_x \|x\|_0 \qquad \text{such that } Ax = y \tag{3.33}$$

Theorem 3.5 (ℓ_1 succed under incoherence)**.** Let $A \in \mathbb{R}^{m \times n}$ be an incoherent matrix with mutual coherence $\mu(A)$, with $diag(A^T A) = I$, $y = Ax_0$, and $\|x_0\|_0 \leq \frac{1}{2\mu(A)}$. Then, x_0 is the unique optimal solution to the ℓ^1 minimization problem.

This theorem can be proven if the three following points are true. Let $x' \in \mathbb{R}^n$ feasible point $\neq 0$, $I = supp(x_0)$, $k = |I|$, we want to show that:

1. $\exists \lambda \in \mathbb{R}^m$ such that
$$\begin{cases} A_I^T \lambda = sign(x_{0_I}) \\ \|A_{I^c}^T \lambda\|_\infty < 1 \end{cases} \tag{3.34}$$
2. $\|x'\|_1 \geq \|x_0\|_1 + (1 - \|A_{I^c}^T \lambda\|_\infty) \|[x' - x_0]_{I^c}\|_1$
3. $[x' - x_0]_{I^c} \neq 0$

If these three points are true, then $\|x'\|_1 > \|x_0\|_1$ and thus x_0 is the unique optimal solution to the ℓ^1 minimization problem.
Let us prove each point:

1. Let $\lambda = A_I (A_I^* A_I)^{-1} sign(x_{0_I})$, then the first condition of 1 is satisfied. For the second condition, we have for $j \in I^c$, then
$$\begin{aligned}
|a_j^* A_I (A_I^* A_I)^{-1} sign(x_{0_I})| &\leq \|A_I^* a_j\|_2 \|(A_I^* A_I)^{-1}\|_{SP} \|sign(x_{0_I})\|_2 \\
&\leq \frac{k\mu(A)}{1 - k\mu(A)} \leq 1
\end{aligned} \tag{3.35}$$
The last inequality is true because $k\mu(A) \leq 1/2$

2. Let $v \in \mathbb{R}^n$ defined by $v_I = sign(x_{0_I})$ and $v_{I^c} = sign([x' - x_0]_{I^c})$. Then if we observe that $v \in \partial\| \cdot \|_1(x_0)$ and that by 1) $A^*\lambda \in \partial\| \cdot \|_1(x_0)$, we have:

$$\begin{aligned} \|x'\|_1 &\geq \|x_0\|_1 + \langle v - A^*\lambda, x' - x_0 \rangle \\ &= \|x_0\|_1 + \langle [v - A^*\lambda]_{I^c}, [x' - x_0]_{I^c} \rangle \\ &= \|x_0\|_1 + \langle v_{I^c}, [x' - x_0]_{I^c} \rangle - \langle A_{I^c}^T\lambda, [x' - x_0]_{I^c} \rangle \\ &\geq \|x_0\|_1 + (1 - \|A_{I^c}^T\lambda\|_\infty)\|[x' - x_0]_{I^c}\|_1 \end{aligned} \tag{3.36}$$

3. By contradiction, suppose $[x' - x_0]_{I^c} = 0$. Then $A(x' - x_0) = 0 = A_I[x' - x_0]_I$. Thus $[x' - x_0]_I = 0$ because A_I is full rank, and thus $x' = x_0$, which is a contradiction.

3.3.1 Constructing incoherent matrices

Theorem 3.6. Let $v \backsim Uni(\mathcal{S}^{n-1})$, then $Pr[|v_n| \geq \varepsilon] \leq \frac{1}{n\varepsilon^2} \xrightarrow[n\to\infty]{} 0$

We can proof this theorem by using the fact that v is of norm 1, and thus $\sum_{i=1}^n v_i^2 = 1$. We can thus say that $\sum_{i=1}^n \mathbb{E}[v_i^2] = 1$ or equivalently $\mathbb{E}[v_n^2] = \frac{1}{n}$. Using the markov inequality $Pr[v_n^2 \geq a] \leq \frac{\mathbb{E}[v_n^2]}{a}$ this gives $Pr[|v_n| \geq \varepsilon] = Pr[v_n^2 \geq \varepsilon^2] \leq \frac{1}{n\varepsilon^2}$.

Theorem 3.7. Let $A = [a_1| \dots |a_n]$, $a_i \backsim Uni(\mathcal{S}^{m-1})$ i.i.d., then $\forall\delta > 0 : \exists C_\delta, \forall m, n$ s.t. $Pr[\mu(A) \geq C_\delta\sqrt{\frac{log(n)}{m}}] \geq 1 - \delta$ or equivalently $\mu(A) \leq C\sqrt{\frac{log(n)}{m}}$ with high probability.

Combining this result with the theorem 3.5, we get that with high probability:

$$\|x_0\|_0 \leq \frac{1}{2C}\sqrt{\frac{m}{log(n)}} \tag{3.37}$$

And we should be able to recover x_0 from $m \geq 4C^2k^2log(n)$ observations.